Project Design Phase-I

Solution Architecture

Date	19 September 2026
Team ID	Vidhya Jain
Project Name	Global Fuel Prices Analysis
Maximum Marks	4 Marks

Solution Architecture

The **Global Fuel Prices Analysis** project uses a data-analytics architecture to compare fuel prices, crude-oil prices, taxes, and subsidies across countries. The dataset covers 84 countries and the period 2020–2026. Data is prepared from CSV/Excel files and analysed and visualised in Tableau. The published Tableau story presents insights through policy, logistics, and research perspectives.

Objectives of the Architecture

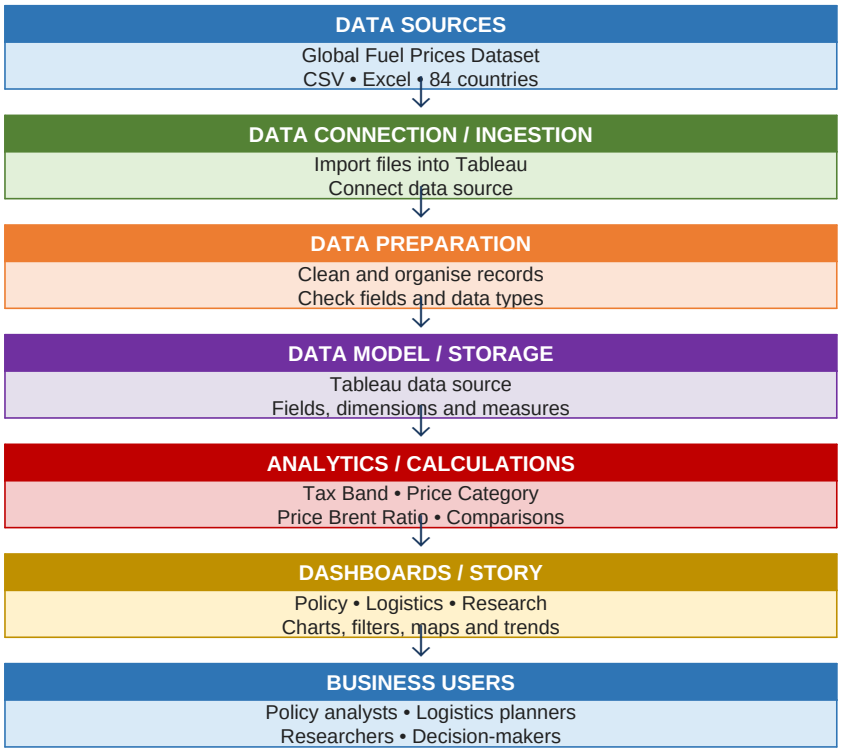
• Bring fuel-price data from CSV and Excel files into one analysis workflow.

• Prepare clean and consistent data for country, region, year, price, tax, and subsidy comparisons.

• Create calculated fields and visualisations to identify price patterns and relationships.

• Present findings in interactive dashboards and a Tableau Public story for decision-making.

Solution Architecture Diagram



Component Description Table

The following table describes the main components used in the project architecture.

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Contains fuel prices, crude-oil prices, tax and subsidy information for country-level analysis.	CSV, Microsoft Excel
Data Preparation	Imports and organises the data; checks fields and data types so the dataset is suitable for analysis.	Excel/CSV, Tableau data connection
Analytics and Calculations	Creates categories and measures to compare price levels and examine the relationship between petrol and Brent crude prices.	Tableau calculated fields: Tax Band, Price Category, Price Brent Ratio
Dashboards and Story	Shows interactive visualisations for policy comparison, logistics planning, and research; supports filtering and country/region comparisons.	Tableau Desktop, Tableau Public
End Users	Uses insights to understand price differences, trends, and tax/subsidy patterns for analysis and planning.	Web browser, Tableau Public

Key Project Outputs

• Three dashboards: Policy, Logistics, and Research.

• Visual comparisons of fuel prices across countries and regions, price trends over 2020–2026, and tax/subsidy patterns.

• Calculated fields: Tax Band, Price Category, and Price Brent Ratio.

• Published Tableau Public story and project files maintained in the GitHub repository.

Note: “Vidhya Jain” has been entered in the Team ID field as requested. If your college expects a numeric or official team code, replace this field with that code.